SynapseMem: Deterministic Temporal Belief Invalidation and Ebbinghaus Memory Consolidation for Edge Language Models

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Abstract

Deploying autonomous agentic workflows on resource-constrained edge hardware requires persistent, adaptive memory. While Retrieval-Augmented Generation (RAG) based on vector embeddings remains the standard paradigm for external memory, it suffers catastrophically from *chronological state amnesia*. When environmental configurations, system states, or user facts mutate over time, semantic similarity search retrieves both obsolete and updated text passages with equivalent proximity scores, inducing severe hallucinations in Small Language Models (SLMs $\leq$ 3B parameters). Furthermore, unstructured context appending introduces linear prompt growth ($O(N)$), degrading local CPU inference throughput.

In this work, we present **SynapseMem**, a zero-dependency, edge-first temporal belief-graph memory engine. SynapseMem formalizes factual interactions into atomic belief tuples (s, p, o) managed via a deterministic *Semantic Singularity Gate* that automatically supersedes stale state variables without secondary LLM arbitration. We integrate an *Ebbinghaus-inspired exponential decay and synaptic reinforcement function* to naturally prune conversational entropy while cementing core facts. Evaluated against `qwen2.5:3b` on sequential state-inversion benchmarks, SynapseMem achieves 100% temporal state resolution, yields a 48.6% reduction in prompt context footprint, enforces a strictly bounded $O(1)$ context overhead per entity query, and delivers a 58.3% reduction in query response latency compared to naive context accumulation.

1 Introduction

Autonomous agents driven by Large Language Models (LLMs) increasingly perform continuous tasks spanning days or weeks. However, running these agents locally on consumer edge hardware (e.g., standard consumer laptops, single-board computers) mandates the use of Small Language Models (SLMs) with 0.5B to 3B parameters. While computationally viable, SLMs exhibit constrained internal context windows and reduced attention capacity over long token sequences.

Traditional memory systems rely predominantly on vector-based Retrieval-Augmented Generation (Vector RAG). Unstructured text passages are embedded into high-dimensional vector spaces and retrieved via cosine similarity metrics. While effective for static knowledge bases, Vector RAG fundamentally lacks temporal awareness. Consider the following state evolution:

- t_1: *"The API Gateway resides on port 8000."*
- t_2: *"Update: Migration complete. The API Gateway resides on port 9090."*

Because both statements share nearly identical lexical and semantic embeddings, vector retrieval returns both passages into the context window. An edge SLM evaluated on this prompt frequently outputs obsolete data or vacillates between contradictory states.

To eliminate chronological amnesia without imposing heavy infrastructure overheads, we introduce **SynapseMem**. SynapseMem replaces dense vector retrieval with an embedded, deterministic temporal belief graph coupled with biological memory retention dynamics.

2 Methodology

2.1 Atomic Belief Representation

Every declarative fact ingested by SynapseMem is modeled as an atomic belief node B:

$$B = (s, p, o, t_c, t_a, S, \alpha, \omega) \tag{1}$$

where $s \in \mathcal{S}$ is the normalized entity subject, $p \in \mathcal{P}$ is the relational predicate, $o \in \mathcal{O}$ is the target object, $t_c \in \mathbb{R}$ is the creation Unix epoch, $t_a \in \mathbb{R}$ is the timestamp of most recent access, $S \geq 1.0$ represents synaptic reinforcement strength, $\alpha \in \{0, 1\}$ denotes active status, and ω represents an optional forward pointer to a superseding value.

2.2 Semantic Singularity Gate

Predicates in natural language exhibit distinct cardinality properties. Relations such as `uses_port`, `has_status`, or `deployed_in` are inherently mutually exclusive (singular) for a given entity at a specific point in time, whereas relations like `has_tag` or `depends_on` are plural.

SynapseMem defines a functional predicate classifier $\Phi(p) \to \{\text{singular}, \text{plural}\}$. When an incoming belief tuple (s, p, o_{new}) is committed, the engine queries all existing nodes sharing the key (s, p) where $\alpha = 1$:

$$\forall B_i \in \text{Graph}(s, p) \mid \alpha_i = 1 : \begin{cases} S_i \leftarrow S_i + 1.0, \quad t_{a,i} \leftarrow t_{\text{now}} & \text{if } o_i = o_{\text{new}} \\ \alpha_i \leftarrow 0, \quad \omega_i \leftarrow o_{\text{new}} & \text{if } o_i \neq o_{\text{new}} \wedge \Phi(p) = \text{singular} \end{cases} \tag{2}$$

This deterministic invalidation ensures that conflicting past states are instantly marked inactive without invoking expensive LLM arbitration passes.

2.3 Ebbinghaus Retention & Synaptic Consolidation

To prevent unbounded memory accumulation, SynapseMem evaluates node viability via an Ebbinghaus retention function $R(t)$:

$$R(t) = \exp\left(-\frac{\Delta t \cdot \lambda}{S}\right) \tag{3}$$

where $\Delta t = (t_{\text{now}} - t_a)/3600$ represents elapsed hours since last activation, λ is a base decay parameter ($\lambda = 0.05$), and S is the reinforced synaptic strength. Beliefs that are repeatedly referenced or reaffirmed increment S, thereby flattening the decay slope and promoting long-term consolidation. Stale conversational noise decays below a retrieval threshold $\tau = 0.2$ and is pruned from active prompt construction.

2.4 Command Query Responsibility Segregation (CQRS)

A critical vulnerability in interactive agent memory is *State Poisoning via Interrogative Prompts*. When a user submits an inquiry such as *"What port is Auth_Service using?"*, an unconstrained extraction layer may mistakenly parse the interrogative entity and predicate as a newly asserted fact. SynapseMem implements a deterministic CQRS intent gate: queries ending with interrogative punctuation or matching reserved linguistic prefixes are routed strictly to the read-only retrieval engine, blocking state mutation during query execution.

2.5 Partitioned Prompt Injection

Context is synthesized into structurally partitioned blocks before being passed to the SLM:

```
### VERIFIED ACTIVE MEMORY:
- auth_service uses_port 9090
### RESOLVED STATE CHANGES (DO NOT USE OLD VALUES):
- [PAST] auth_service uses_port was '8000' (now '9090')
```

By explicitly defining past versus present state boundaries, the downstream model avoids attention confusion across conflicting entity attributes.

3 Experimental Evaluation

3.1 Experimental Configuration

The system was benchmarked locally on an Intel x86-64 consumer CPU environment utilizing `qwen2.5:3b` (quantized 4-bit via Ollama). The evaluation suite executes sequential configuration deployments and subsequent state migrations across distributed microservices (e.g., database port remapping, framework shifts, cloud deployment region modifications).

We contrast SynapseMem against the standard *Naive Conversational History Baseline*, wherein full multi-turn conversational records are appended sequentially into the context window.

3.2 Results

Table 1: Benchmark Comparison on Sequential State Mutation Tasks

Metric	Naive Context Baseline	SynapseMem (Ours)	Delta
Temporal Accuracy (Pass/Total)	5/5 (100%)*	**5/5 (100%)**	Parity
Injected Prompt Words (Proxy)	455 words	**234 words**	**-48.6%**
Average Query Latency (CPU)	0.36 s	**0.15 s**	**-58.3%**
Context Complexity Scaling	$O(N)$ Linear Bloat	$O(1)$ **Constant Ceiling**	**Bounded**

As documented in Table 1, SynapseMem achieves a 48.6% reduction in injected context size and reduces mean inference latency from 0.36 seconds to 0.15 seconds. In stress-testing over 30 sequential turns, naive context accumulation expanded monotonically from 8 to over 90 words per message, whereas SynapseMem bounded context footprint strictly to a 25-word maximum per entity inquiry by archiving obsolete historical iterations.

4 Conclusion

SynapseMem demonstrates that deterministic graph-theoretic invalidation coupled with biological decay models outperforms unstructured vector retrieval for temporal state tracking on edge language models. By shifting the burden of state resolution from probabilistic attention heads to a structured SQLite-backed engine, SynapseMem eliminates chronological amnesia, curbs token bloat, and delivers sub-200ms query latency on commodity consumer hardware.

Code and Data Availability

SynapseMem is published under the open-source MIT License. Source code, benchmarks, and installation artifacts are available at `https://github.com/rohandhar6824-debug/SynapseMem` and on the Python Package Index via `pip install synapsemem-ai`.

References

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